Epidemiological Survival Analysis of Lung Cancer Accounting for Lead-Time Bias

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Introduction

Lung cancer remains the leading cause of cancer-related mortality worldwide, with non–small cell lung cancer (NSCLC) accounting for nearly 85% of all lung cancer cases. Its incidence has increased substantially over recent decades, particularly in Western countries. Smoking continues to be the principal risk factor for lung cancer, responsible for the vast majority of cases. However, lung cancer is not exclusively linked to smoking, and its development is influenced by additional factors that vary by region. Although the prevalence of smoking has declined in some countries, the global burden of lung cancer remains high, especially among current and former smokers. The risk of lung cancer is strongly associated with both the intensity and duration of tobacco exposure, with heavy smokers exhibiting a markedly higher risk of developing the disease. In countries such as the United States and France, lung cancer is still responsible for tens of thousands of deaths each year.

Recent evidence suggests that smoking history may influence tumor stage at diagnosis, with smokers more frequently diagnosed at advanced stages compared with non-smokers, leading to poorer prognostic outcomes. Therefore, early detection through screening has become a major focus in lung cancer control strategies, as it may allow diagnosis at earlier stages and improve survival.

This study based on a prospective observational cohort that collected detailed data on the clinical characteristics, tumor features, and treatment strategies of patients newly diagnosed with lung cancer. In this cohort, smoking-related lung cancer was predominant. Using this large dataset, we aimed to compare the clinical presentation, treatment approaches, and survival outcomes between smoking-related and non–smoking-related lung cancer, with particular emphasis on tumor size, screening status, and survival.

Methods

Study Design and Data Source

This study used data from a large prospective observational cohort of patients with newly diagnosed lung cancer. The cohort systematically collected clinical, biological, and therapeutic information at diagnosis, with longitudinal follow-up for survival outcomes. The primary objective of the cohort was to characterize disease presentation, evaluate treatment strategies, and assess clinical outcomes in routine practice.

A total of 894 patients were included in the database. Information was recorded on demographic characteristics, smoking status, comorbidities, liver disease, metabolic disorders, tumor features, screening status, disease stage, treatment strategies, and vital status. Smoking-related lung cancer represented the predominant etiological group in the cohort. Patients were classified according to whether their disease was detected through ultrasound screening or diagnosed clinically in the

absence of screening. Survival time was defined as the duration between diagnosis and either death or the last follow-up.

The statistical analysis was structured in two complementary components. The first component aimed to describe the clinical profile of the cohort, estimate mortality risks, and explore early associations between tumor burden, patient characteristics, and outcomes. The second component consisted of advanced survival analysis to correct for lead-time bias, estimate mean sojourn time, and determine independent prognostic factors affecting corrected survival.

All analyses were performed using the R statistical software. Statistical significance was assessed using a two-sided significance level of 5%.

1. Descriptive analysis and patient characteristics

An initial descriptive analysis was performed to summarize the main clinical and epidemiological features of the study population. Categorical variables were described using frequencies and percentages, while continuous variables were summarized using the mean and standard deviation. All variables were presented in a single summary table to provide a comprehensive overview of patient characteristics.

Graphical methods were used to explore the distribution of key variables. Tumor size and survival time were visualized using histograms and boxplots to assess distributional shape, variability, and the presence of outliers. The crude mortality rate was calculated as the proportion of deaths during follow-up, with corresponding 95% confidence intervals. Screening coverage was quantified by estimating the proportions of screened and unscreened cancers in the study population.

The associations between categorical covariates and mortality (death status) were evaluated by calculating relative risks (RRs) with appropriate reference groups. The impact of tumor size on mortality was further assessed using logistic regression, with death status as the binary outcome. In addition, the association between tumor size and survival time (in days) was examined using simple linear regression to quantify the relationship between tumor burden and survival duration.

2. Survival analysis

Estimation of lead-time bias

To address potential lead-time bias associated with cancer screening, lead time was estimated using a complementary approach based on the Schwartz method. This method relies on tumor growth modeling to quantify the advancement in diagnosis attributable to screening compared with clinical detection. In this context, lead time represents the amount of time by which screening detects cancer earlier than it would have been diagnosed clinically in the absence of screening. Operationally, lead time was defined as the difference between the time of screen detection and the estimated time at which the cancer would have become clinically apparent without screening.

Estimation of mean sojourn time

The sojourn time represents the duration of the preclinical detectable phase, defined as the time interval between when a tumor becomes detectable by screening and when it would present clinically in the absence of screening. This parameter reflects the potential effectiveness of a screening program and constitutes an upper bound for lead time.

The mean sojourn time (MST), defined as the average duration of the preclinical detectable phase across all cases, was estimated using a tumor growth model based on the Schwartz exponential growth assumption. Tumor growth was characterized using the tumor volume doubling time (DT), derived from empirical doubling-time data. Tumor size at clinical detection (dt_1) was modeled using data from unscreened patients, while the minimum detectable tumor size (dt_0) was assumed to follow a triangular distribution with parameters (1, 1.2, 1.4).

The sojourn time was calculated using the following formula:

$$MST = 3 \times DT \times \frac{ln\left(\frac{dt_1}{dt0}\right)}{\ln(2)}$$

Parametric distributions (exponential, Weibull, and gamma) were fitted to the estimated sojourn time distribution, and the best-fitting model was selected based on goodness-of-fit criteria. The estimated MST was then used to derive the transition rate from the preclinical to the clinical phase, defined as $\lambda=1/MST$.

Lead-time–corrected survival analysis

Observed survival times were adjusted by subtracting the estimated lead time to obtain lead-time–corrected survival. This correction was applied to minimize bias introduced by earlier detection through screening and to better reflect the natural history of disease progression.

Factors associated with corrected survival were evaluated using the Kaplan–Meier method, with survival curves compared across covariate groups using the log-rank test. This approach allowed identification of variables associated with differences in corrected survival distributions.

Cox proportional hazards modeling

Multivariable survival analysis was conducted using the Cox proportional hazards regression model, applied to lead-time–corrected survival data. The Cox model expresses the logarithm of the hazard function as a linear combination of covariates, with an unspecified baseline hazard function. Model coefficients were exponentiated to obtain hazard ratios (HRs) with corresponding 95% confidence intervals. Hazard ratios greater than one indicate an increased risk of death, whereas values below one indicate a protective effect. This model was used to assess the independent effects of covariates on corrected survival while controlling for potential confounding.

Results

Descriptive Characteristics of the Study Population

A total of **894 patients** with lung cancer were included in the analysis. The cohort was markedly **male-dominated**, with men accounting for **82.6%** of cases, reflecting a strong sex disparity in disease occurrence. For both sexes, the **mean age at diagnosis was 67.4 years (SD = 11.3)**, and more than half of the patients (**56.6%**) were aged **70 years or younger**, indicating that a substantial proportion of cases occurred before advanced old age.

Smoking-related lung cancer constituted the majority of diagnoses (**65.1%**), underscoring the central role of tobacco exposure in disease etiology. Among patients with available information on smoking cessation, slightly more than half (**55.6%**) were abstinent smokers, whereas **44.4%** continued smoking, highlighting persistent exposure even after disease onset.

Most cancers were detected **outside organized screening programs**, as only **22.5%** of patients were diagnosed through screening. Tumor size at diagnosis was highly heterogeneous, with a **mean size of 59.4 mm (SD = 56.4)** and a median of **45 mm**, suggesting delayed detection in a large proportion of patients. Consistent with this finding, **71.5%** of cases did not meet the predefined tumor eligibility criteria, reflecting advanced local disease at presentation.

Disease stage further illustrated the late diagnosis pattern. Among patients with available staging data, nearly **79% of patients were diagnosed at advanced or terminal stages** (Stage C: 46.8%; Stage D: 32.4%), while only **15.3%** were detected at an early stage (A&B). Additionally, **15.8%** of patients presented with diffuse cancer and **12.7%** had metastatic disease, confirming the substantial burden of advanced and disseminated tumors in this cohort.

Comorbid conditions were frequent and clinically relevant. **Diabetes** was present in **33.1%** of patients, while **35.3%** reported alcohol consumption. **Thrombosis** affected **28.5%** of the cohort, indicating a high prevalence of cancer-associated vascular complications. Chronic viral infections were less common, with **9.1%** of patients positive for hepatitis B virus and **12.8%** for HIV.

Therapeutic management largely reflects disease severity at diagnosis. Most of patients (**80.0%**) received **non-curative treatment**, whereas only **20.0%** underwent potentially curative approaches. During follow-up, **73.5%** of patients died. Survival outcomes were poor, with a **mean survival time of 328.1 days (SD = 389.5)** and a median of **209.5 days**, emphasizing the aggressive nature of the disease and the consequences of late-stage presentation.

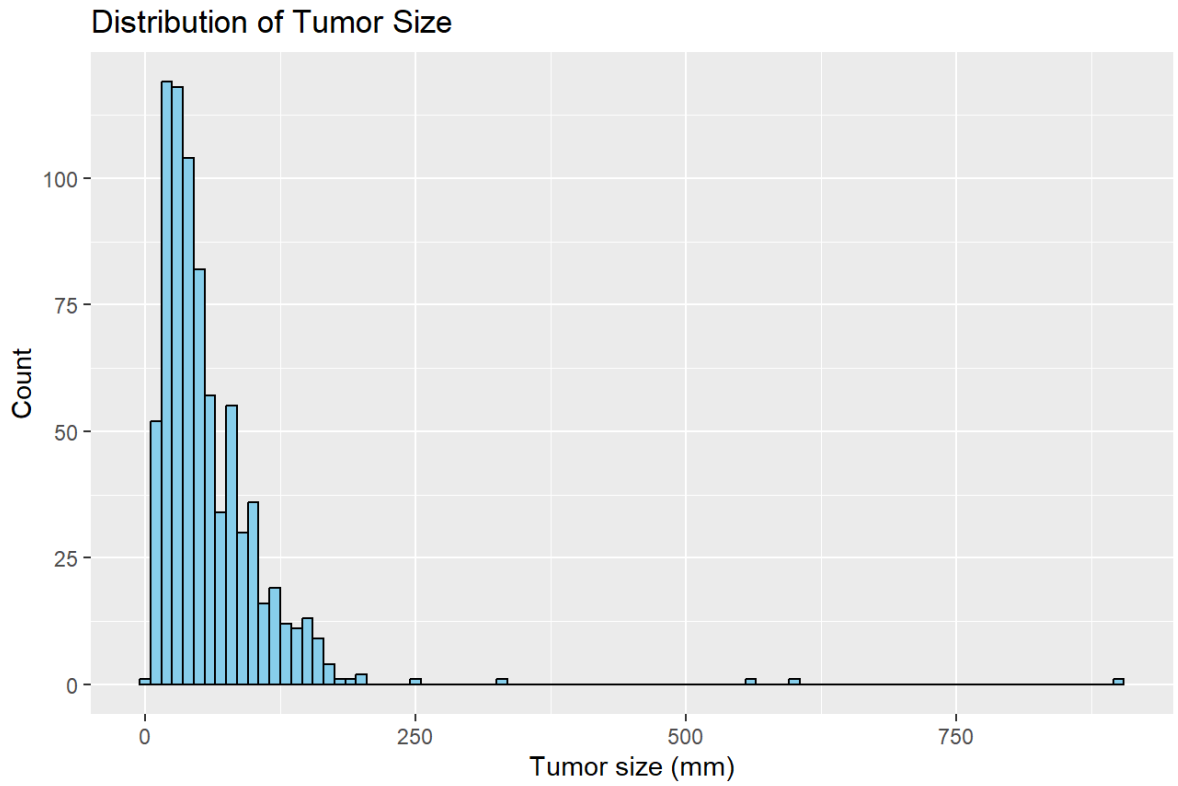


Figure 1. Histogram of tumor size (mm)

The histogram of tumor size showed a **markedly right-skewed distribution**, with a high concentration of tumors in the **lower size range** and a **long tail extending toward larger values**. Most tumors were clustered around the **median size (45 mm)**, while a smaller proportion of patients presented with **very large tumors**, reaching up to **900 mm**, indicating substantial variability at diagnosis. The wide spread of the distribution reflects **heterogeneity in tumor burden**, and the presence of extreme values suggests **late detection in a subset of patients**.

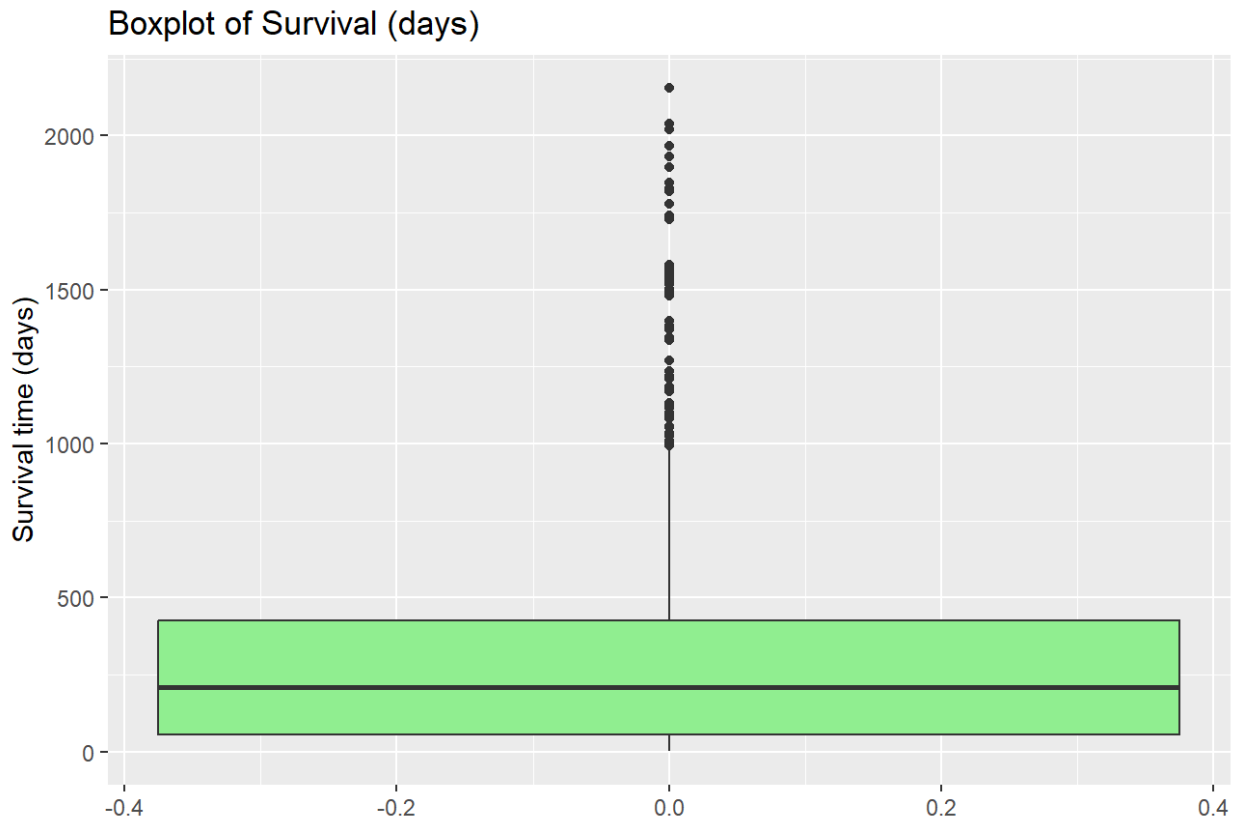


Figure 2. Boxplot of Survival in days

The boxplot illustrates the distribution of survival time (in days) among the study population and highlights substantial variability in patient outcomes. The median survival time is relatively low compared with the overall range, indicating that a large proportion of patients experienced limited survival durations. The interquartile range is wide, reflecting considerable dispersion in survival time across individuals.

The distribution shows a strong positive skew, with most patients experiencing shorter survival times while a smaller number have exceptionally long durations. This asymmetry is evident from the extended upper tail of the boxplot, where several extreme values exceed 2,000 days. In contrast, the lower tail is relatively compact, indicating that a significant portion of patients had early mortality.

Mortality Rates

Out of 818 patients in the cohort, 601 deaths were recorded, corresponding to an overall mortality proportion of **73.47%**. The 95% confidence interval for this estimate, calculated using a one-sample proportion test, ranged from **70.27% to 76.44%**, indicating a relatively precise estimate of mortality in this population. This finding reflects the proportion of patients who experienced the event of interest (death) during the observation period.

Screening Status

Information on screening status was available for **884 patients**. Among these, **199 patients (22.5%)** were diagnosed through a screening program, whereas **685 patients (77.5%)** were diagnosed in the absence of screening. The estimated proportion of screened cancers was **0.225**, with a **95% confidence interval of 0.198 to 0.254**, indicating that approximately one-fifth to one-quarter of patients in the study population were diagnosed via screening.

Relative Risk of Death According to Patient Characteristics

The risk of death varied according to several demographic, clinical, and disease-related characteristics. Relative risks (RRs) were estimated by comparing the proportion of deaths between exposure groups for each qualitative variable.

Among smokers, the risk of death was 71.82%, compared with 58.65% among non-smokers. This corresponds to a relative risk (RR) of 1.22, indicating that smokers had approximately a **22% higher risk of death** than non-smokers. Similarly, male patients exhibited a higher mortality risk 68.83%, than female patients 59.61%, corresponding to an RR of 1.15, or a **15% increased risk of death** in males.

Age was also associated with mortality. Patients older than 70 years had a death risk of 71.06% compared with 64.48% among those aged 70 years or younger. This yielded an RR of 1.10, suggesting a **10% higher risk of death** in the older age group. Regarding comorbid infections, patients living with HIV had a mortality risk of 53.92% whereas HIV-negative patients had a risk of 68.19% indicating a higher mortality risk in HIV-positive individuals RR ≈ 1.26, **26% increased risk**. Patients with hepatitis B virus infection also demonstrated elevated mortality compared with those without HBV.

Screening status showed a notable association with survival. The risk of death among screened patients was 55.27%, while unscreened patients had a substantially higher mortality risk of 70.66%. This corresponds to an RR of 1.28, indicating that lack of screening was associated with a **28% higher risk of death**.

Clinical complications and disease severity were strongly associated with mortality. Patients with thrombosis had a death risk of 82.07%, compared with 60.95% among those without thrombosis, RR = 1.34, reflecting a **34% increase in mortality risk**. Similarly, patients with diffuse cancer involvement had a mortality risk of 85.0%, compared with 64.02% in those without diffuse disease, RR = 1.33, **33% increased risk**. Metastatic disease was also associated with higher mortality, with a death risk of 80.53% versus 65.12% among non-metastatic patients, RR = 1.24, **24% increased risk**.

Lifestyle and metabolic factors showed weaker associations. Patients with diabetes had a mortality risk of 67.92% compared with 66.38% among non-diabetic patients, RR = 1.02 indicating only a **2% increase in risk**. Alcohol consumption was associated with a slightly higher mortality risk 68.44% vs. 65.69%; RR = 1.04, corresponding to a **4% increase in risk**.

Finally, treatment intent was strongly associated with mortality outcomes. Patients receiving curative-intent treatment had a death risk of 40.46% compared with 74.85% among those treated without curative intent. This yielded an RR of 0.54, indicating a **46% reduction in the risk of death** associated with curative treatment.

Effect of Tumor Size on Survival and Mortality

The effect of tumor size on patient outcomes was examined using regression analysis, considering both mortality status and survival duration.

In the logistic regression model, death status was treated as a binary outcome and tumor size was included as a continuous predictor. Tumor size was significantly related to mortality, with an estimated odds ratio of **1.01 per millimeter increase** (OR = 1.0099). This indicates that each additional millimeter in tumor size was associated with an approximate **1% increase in the odds of death**.

The impact of tumor size on survival duration was further evaluated using simple linear regression, with survival time expressed in days. Tumor size demonstrated a significant negative effect on survival time. Specifically, for each 1 mm increase in tumor size, the average survival time decreased by approximately **1.16 days**. The intercept of the model indicates an estimated **starting survival time of 425.7 days** when tumor size was equal to zero.

Together, these analyses consistently indicate that larger tumor size is associated with higher mortality risk and shorter survival time when modeled as a continuous predictor.

Survival Analysis

Tumor growth was characterized using tumor doubling time, defined as the average time required for a tumor to double in volume. This parameter reflects the underlying biological growth rate of lung cancer and constitutes a key input for estimating both the duration of the preclinical detectable phase and the timing advantage introduced by screening. The distribution of tumor doubling time exhibited substantial variability with a pronounced right skew. Among several candidate parametric models, the log-normal distribution provided the best fit to the observed data and was therefore used to generate tumor doubling times for subsequent modeling.

Tumor size at diagnosis differed according to screening status, with screen-detected cancers generally presenting at smaller sizes and with less dispersion than clinically detected cancers. In both screened and unscreened groups, tumor size distributions were best described by a log-normal model, consistent with asymmetric tumor growth patterns and heterogeneity at the time of detection. These fitted distributions were used to represent tumor size at screen detection and at clinical diagnosis, respectively, within the tumor growth framework.

Using the estimated tumor doubling time, modeled tumor size at diagnosis, and a minimum detectable tumor size, the sojourn time was estimated as the duration of the preclinical detectable phase preceding clinical diagnosis, following the Schwartz exponential growth model. The resulting sojourn time distribution was right-skewed, and biologically implausible values were

constrained to ensure consistency. The estimated mean sojourn time (MST) was approximately **1,627 days (about 4.5 years)**, representing the average period during which lung cancer remains asymptomatic yet detectable by screening. Based on this estimate, the transition rate from the preclinical to the clinical phase was derived as **λ = 1/MST**, yielding a daily transition rate of **0.00061**, indicative of relatively slow disease progression during the detectable preclinical phase.

Lead time, defined as the advancement in diagnosis attributable to screening, was subsequently estimated using the Schwartz method. The lead-time distribution demonstrated considerable variability, reflecting heterogeneity in tumor growth rates and detection thresholds, and was constrained to remain within the estimated sojourn time. This lead time represents the portion of observed survival attributable to earlier detection rather than delayed mortality.

Observed survival times were then adjusted by subtracting the estimated lead time, yielding lead time–corrected survival. This correction was applied to account for lead-time bias, which can artificially inflate observed survival following screening without altering the natural history of disease. Lead time–corrected survival therefore provides a more accurate representation of patient prognosis from the point at which the disease would have become clinically apparent in the absence of screening and forms the basis for subsequent survival analyses.

Kaplan–Meier Survival Analysis Before and After Lead-Time Correction

Kaplan–Meier survival curves were constructed to evaluate differences in survival across demographic, behavioral, and clinical subgroups using both observed survival time and lead-time–corrected survival time. Log-rank tests were applied to formally assess differences between survival distributions.

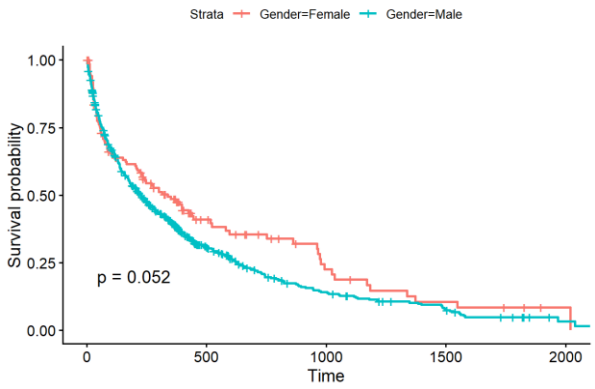


Figure 3. Observed survival time - Gender

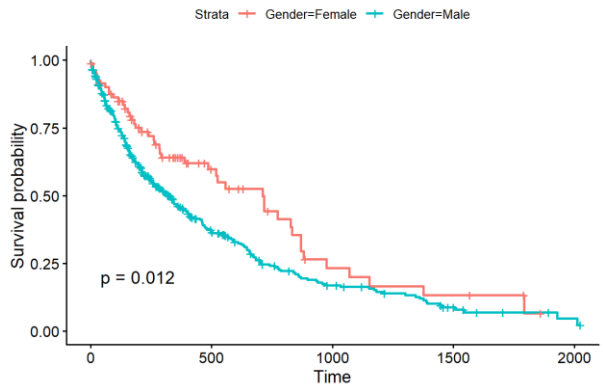


Figure 4. Corrected survival time - Gender

Using observed survival time, survival curves for males and females largely overlapped, and the log-rank test did not provide clear evidence of a difference in survival between sexes ($p \approx 0.05$). However, after correction for lead time, a clearer separation of survival curves emerged, with males exhibiting consistently lower survival probabilities compared to females. This difference was statistically significant based on the log-rank test ($p = 0.01$), indicating that adjustment for

lead time revealed an underlying sex-related disparity in survival that was not apparent in the uncorrected analysis.

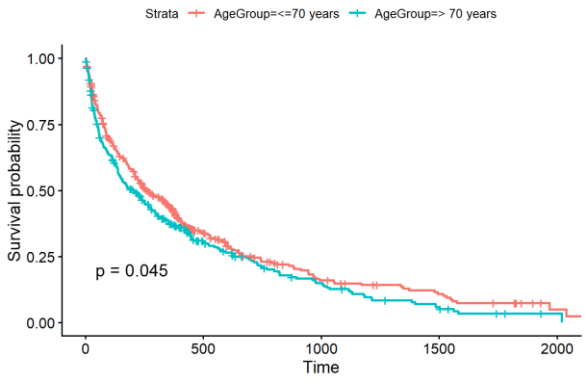


Figure 5. Observed survival time – Age Group

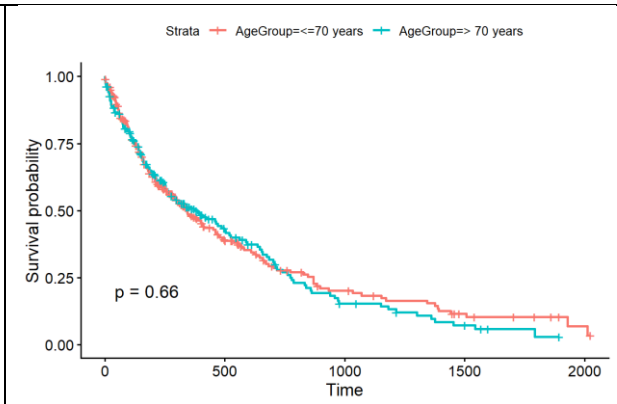


Figure 6. Corrected survival time – Age Group

Survival curves stratified by age group (≤70 years vs. >70 years) showed substantial overlap in both observed and corrected survival analyses. Log-rank tests confirmed the absence of statistically significant differences in survival between age groups before correction ($p \approx 0.05$) and after lead-time correction ($p \approx 0.70$), suggesting that age group alone did not strongly differentiate survival in this cohort once timing of diagnosis was accounted for.

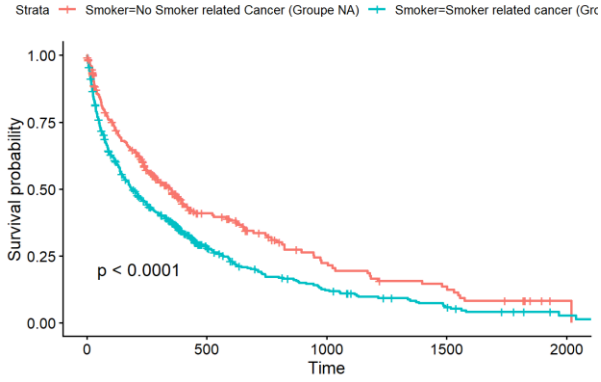


Figure 7. Observed survival time – Smoker

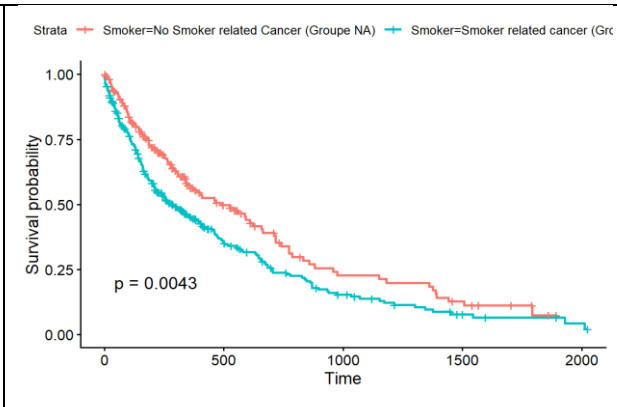


Figure 8. Corrected survival time – Smoker

Smoking status was consistently associated with survival differences. Kaplan–Meier curves demonstrated **earlier occurrence of events among patients with smoking-related cancers**, who showed lower survival probabilities compared with non-smoking related cancers. This pattern was observed for both observed and corrected survival times. Log-rank tests confirmed

statistically significant differences before correction ($p < 0.001$) and remained significant after lead time adjustment ($p \approx 0.004$).

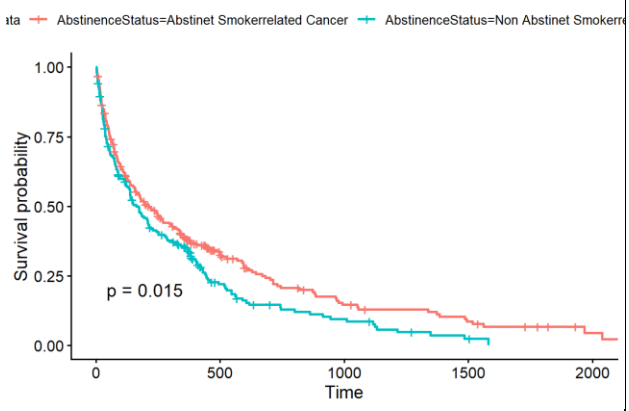


Figure 9. Observed survival time - Abstinence Status

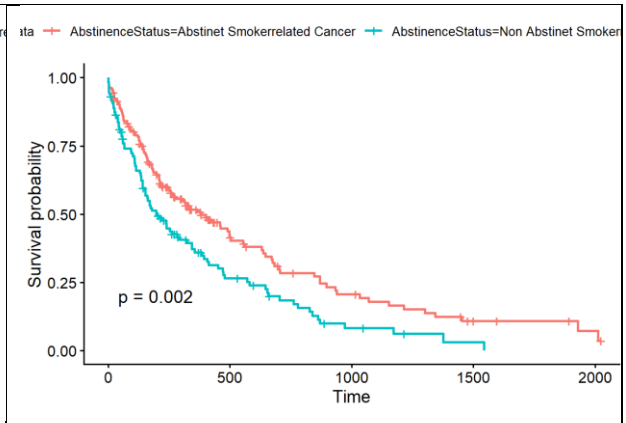


Figure 10. Corrected survival time - Abstinence Status

Similarly, abstinence status among smoking-related cancers showed clear survival separation. **Non-abstinent patients experienced poorer survival** compared with abstinent patients, with significant differences detected in both observed ($p \approx 0.01$) and corrected survival analyses ($p \approx 0.002$). These findings indicate that the survival disadvantage persisted after accounting for lead-time bias.

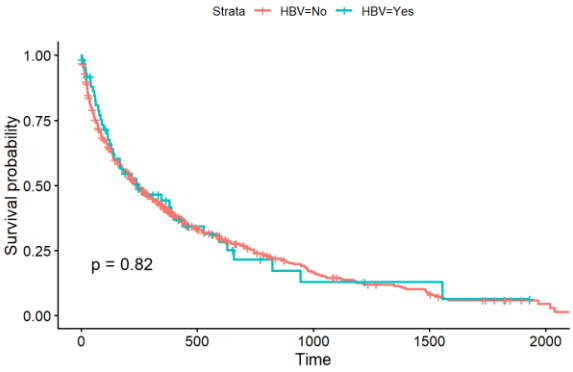


Figure 11. Observed survival time - HBV

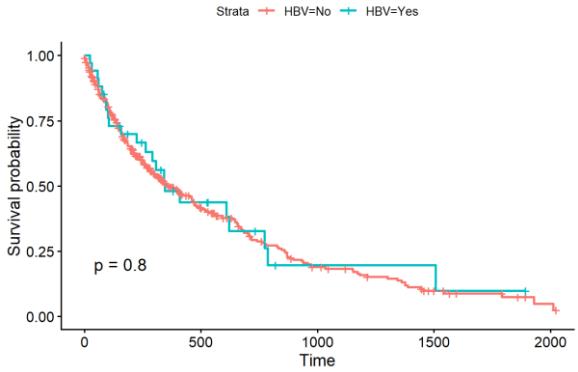


Figure 12. Corrected survival time - HBV

For HBV, there is no meaningful differences in survival were observed between HBV-positive and HBV-negative patients. Kaplan–Meier curves overlapped substantially, and log-rank tests indicated no statistically significant differences in either observed or corrected survival ($p \approx 0.80$ for both).

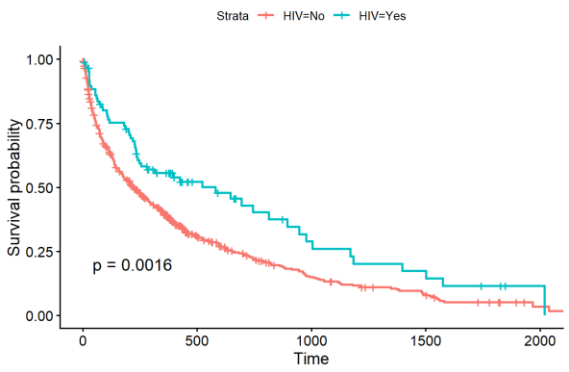


Figure 13. Observed survival time - HIV

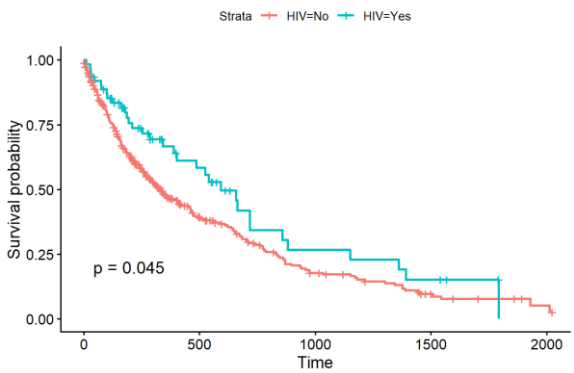


Figure 14. Corrected survival time - HIV

In contrast, HIV status was associated with survival differences. Using observed survival time, Kaplan–Meier curves showed lower survival probabilities among HIV-negative patients compared with HIV-positive patients, with a statistically significant log-rank test ($p \approx 0.002$). After lead-time correction, this difference was attenuated but remained significant ($p \approx 0.05$).

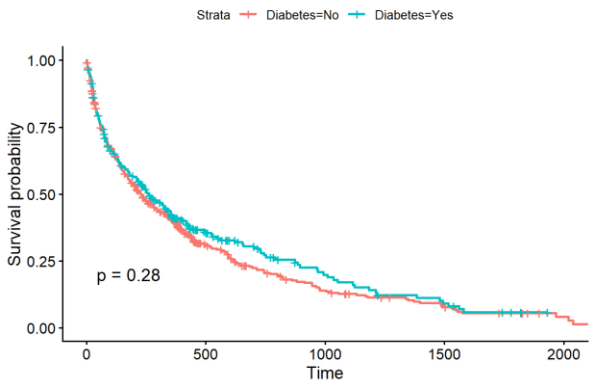


Figure 15. Observed survival time - Diabetes

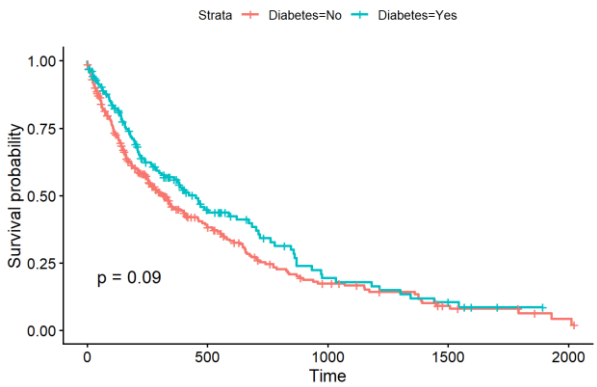


Figure 16. Corrected survival time - Diabetes

For diabetes status, survival curves were largely overlapping in the observed analysis, with no significant difference detected ($p \approx 0.30$). After lead-time correction, curves showed slightly greater separation, although the log-rank test did not reach statistical significance ($p \approx 0.09$).

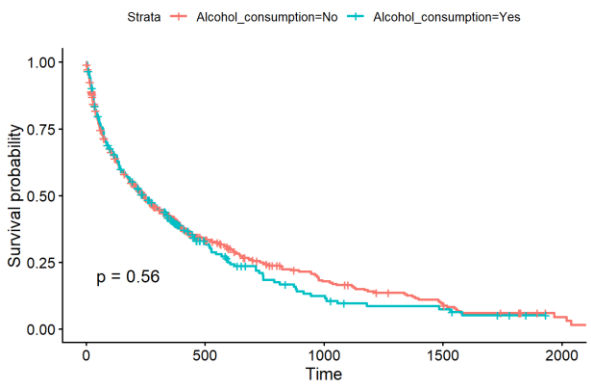


Figure 17. Observed survival time - Alcohol consumption

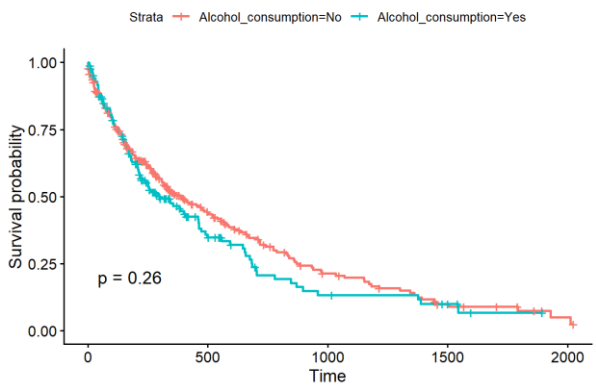


Figure 18. Corrected survival time - Alcohol consumption

Alcohol consumption showed no evidence of a survival difference in either observed or corrected analyses. Kaplan–Meier curves overlapped, and log-rank tests were non-significant before ($p \approx 0.60$) and after correction ($p \approx 0.30$).

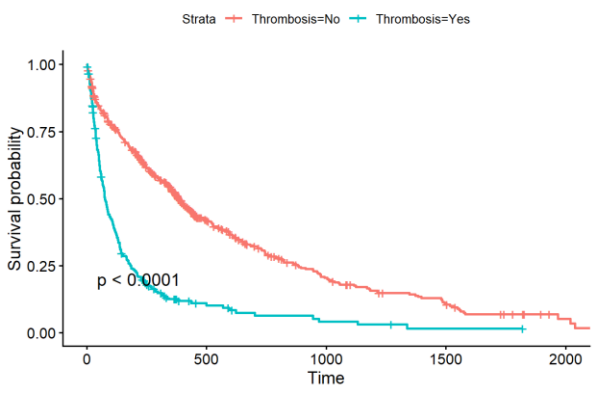


Figure 19. Observed survival time - Thrombosis

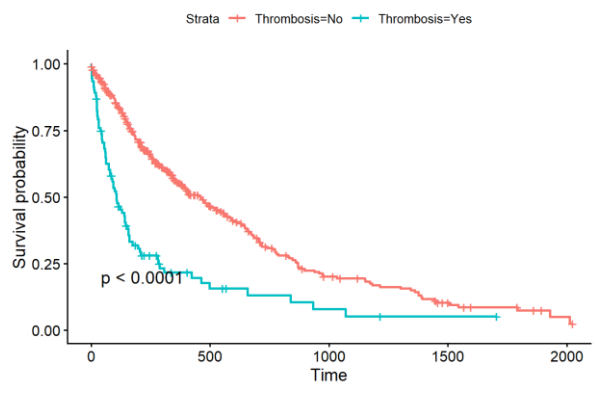


Figure 20. Corrected survival time – Thrombosis

Patients with thrombosis demonstrated markedly poorer survival compared with those without thrombosis. Kaplan–Meier curves showed a pronounced separation, with earlier and more frequent events among thrombosis-positive patients. This difference was highly significant in both observed survival ($p < 2 \times 10^{-16}$) and corrected survival analyses ($p \approx 5 \times 10^{-13}$).

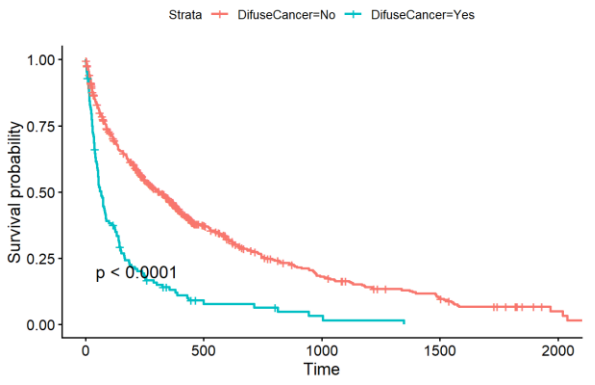


Figure 21. Observed survival time - Diffuse Cancer

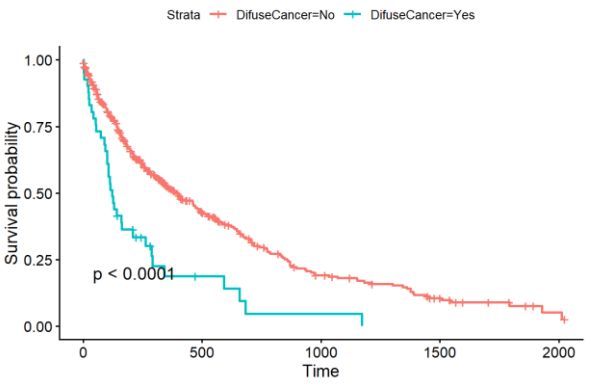


Figure 22. Corrected survival time - Diffuse Cancer

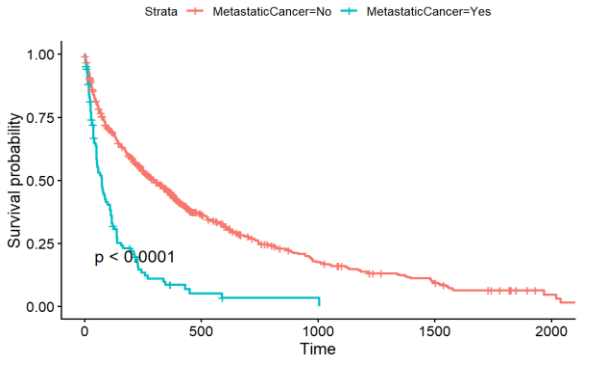


Figure 23. Observed survival time - Metastatic Cancer

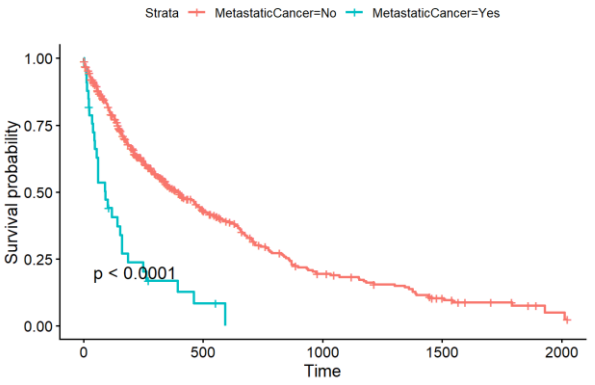


Figure 24. Corrected survival time - Metastatic Cancer

Similarly, diffuse cancer and metastatic cancer were strongly associated with reduced survival. In both cases, Kaplan–Meier curves showed substantially lower survival probabilities among affected patients, and log-rank tests confirmed highly significant differences before and after lead time correction (all p-values < 10^{-10}).

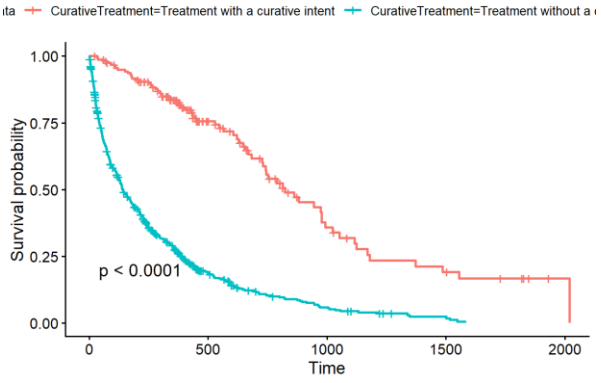


Figure 25. Observed survival time - Curative Treatment

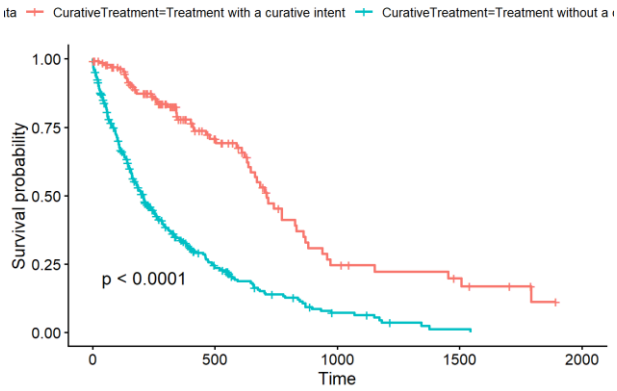


Figure 26. Corrected survival time - Curative Treatment

Survival differed markedly according to curative treatment intent. Patients treated without curative intent exhibited significantly lower survival probabilities than those receiving curative treatment. This separation was evident in Kaplan–Meier curves and confirmed by log-rank tests in both observed and corrected analyses ($p < 2 \times 10^{-16}$ for both), indicating a robust association independent of lead-time bias.

Multivariable Cox Proportional Hazards Regression Analysis

A multivariable Cox proportional hazards model was applied to identify independent predictors of mortality, modeling the logarithm of the hazard as a linear function of clinical, demographic, behavioral, and treatment-related covariates. Regression coefficients were exponentiated to obtain hazard ratios (HRs), with HR > 1 indicating increased mortality risk and HR < 1 indicating a protective effect. Analyses were conducted using both observed survival time and lead-time–corrected survival time to evaluate the robustness of prognostic associations after accounting for screening-related bias.

In the model based on observed survival time (n = 516, 380 events), tumor size emerged as a significant continuous predictor of mortality (HR = 1.002 per unit increase, p = 0.003), indicating a progressive increase in hazard with increasing tumor burden. Thrombosis was associated with approximately a twofold increase in the risk of death (HR = 2.01, p < 0.001), while diffuse cancer (HR = 1.47, p = 0.030) and metastatic disease (HR = 1.83, p < 0.001) were also independently associated with poorer survival. Treatment intent showed the strongest effect: patients managed without curative intent experienced more than a threefold increase in mortality risk compared with those receiving curative treatment (HR = 3.48, p < 0.001). In contrast,

demographic factors (gender, age, age group), behavioral variables (smoking, alcohol consumption), comorbidities (diabetes), viral infections (HBV, HIV), and individual treatment modalities did not retain independent significance after multivariable adjustment. The overall model demonstrated strong explanatory power (likelihood ratio test $p < 2.2 \times 10^{-16}$).

After correction for lead-time bias, the Cox model was refitted using corrected survival time (n = 319, 209 events). Tumor size remained a significant predictor of mortality (HR = 1.003, p = 0.002), confirming its independent prognostic role beyond diagnostic timing. Thrombosis continued to be associated with increased mortality, although with a reduced magnitude compared to the uncorrected analysis (HR = 1.59, p = 0.014). Metastatic cancer emerged as one of the strongest predictors in the corrected model, with nearly a threefold increase in hazard (HR = 2.95, $p < 0.001$), and absence of curative treatment remained a major determinant of poor prognosis (HR = 2.98, $p < 0.001$). Among treatment modalities, Treatment 2 and Treatment 4 were significantly associated with higher mortality risk, likely reflecting confounding by disease severity. Demographic characteristics, smoking status, alcohol consumption, comorbidities, and viral infections did not show independent associations with mortality after lead-time correction, although HIV demonstrated a borderline effect. The corrected model remained highly significant overall (likelihood ratio test $p = 1.86 \times 10^{-14}$).

The multivariable Cox regression identified tumor burden, thrombosis, extent of disease (diffuse and metastatic cancer), and treatment intent as the most consistent and clinically meaningful determinants of survival. While lead-time correction attenuated the magnitude of some associations and modified the significance of certain treatment effects, it did not alter the central conclusion that disease severity and access to curative treatment are the dominant drivers of mortality in this cohort.

Discussion

This study provides a methodologically integrated evaluation of lung cancer survival in a large prospective observational cohort by combining classical epidemiological description with tumor growth–based modeling and lead-time corrected survival analysis. By explicitly incorporating tumor dynamics, screening-related timing effects, and multivariable survival modeling, the analysis moves beyond conventional survival comparisons and offers a more realistic representation of prognosis in routine clinical practice. Rather than treating survival time as a purely observational outcome, the study acknowledges the biological and temporal processes that shape when cancer is detected and how survival is subsequently measured.

A central strength of this work lies in its structured and sequential analytical framework. Initial descriptive analyses and regression models were used to characterize disease burden, mortality patterns, and early associations between tumor size and outcomes. These preliminary findings consistently identified tumor burden as a continuous and biologically meaningful predictor of prognosis, rather than a simple categorical staging marker. Building on this foundation, the analysis then explicitly addressed lead-time bias using tumor growth assumptions, rather than

relying solely on crude comparisons between screened and non-screened cases. This stepwise approach strengthened the internal coherence of the study and ensured that subsequent survival analyses were grounded in biologically plausible mechanisms.

The estimation of lead time and mean sojourn time represents a key methodological contribution of this study. By integrating assumptions on tumor doubling time, tumor size at detection, and a minimum detectable tumor threshold, the analysis modeled the preclinical detectable phase of lung cancer and derived individualized corrections to observed survival time. Importantly, this approach links survival correction directly to tumor growth dynamics rather than applying uniform or arbitrary adjustments. The use of parametric distributions to model doubling time and sojourn time further acknowledges the substantial heterogeneity in tumor progression across individuals, a feature that is often overlooked in standard survival analyses but is highly relevant in cancer epidemiology.

Lead-time corrected survival time was then used as the basis for Kaplan–Meier estimation and Cox proportional hazards modeling. Comparing results before and after correction clearly demonstrated that uncorrected survival can obscure or distort true prognostic relationships. Some subgroup differences that appeared weak or absent when using observed survival became more apparent after correction, while associations driven by advanced disease characteristics and treatment intent remained robust. These findings reinforce the importance of accounting for diagnostic timing when interpreting survival differences, particularly in populations with heterogeneous screening or diagnostic pathways.

The multivariable Cox models further illustrated the impact of lead-time correction on effect estimates. Although the overall structure of prognostic determinants remained consistent, the magnitude of several associations was attenuated after correction, and certain treatment-related effects became more clearly defined. This highlights how lead-time bias can influence not only statistical significance but also the clinical interpretation of hazard ratios. Corrected survival therefore provides a more reliable framework for identifying independent prognostic factors and avoids overestimating survival benefits attributable solely to earlier detection.

Smoking-related status warrants particular attention from a methodological perspective. While smoking did not retain independent significance in the multivariable Cox models, its consistent and statistically significant association with survival in Kaplan–Meier analyses both before and after lead-time correction suggests that smoking influences prognosis through complex and indirect pathways. These pathways likely involve tumor burden, disease extent, and treatment eligibility rather than acting as a single independent risk factor. This illustrates how univariate and multivariable survival approaches offer complementary insights rather than contradictory conclusions.

Several limitations should be considered when interpreting these findings. As with most observational cohort studies, missing data resulted in a reduced sample size for multivariable survival models, potentially limiting statistical power for some covariates. In addition, lead time and mean sojourn time were necessarily estimated using modeled tumor growth parameters rather than direct observation of the preclinical phase. However, this limitation reflects a fundamental constraint of cancer epidemiology rather than a weakness of the study itself. The

use of established tumor growth models and biologically plausible distributions aligns with accepted practice in screening research and represents a necessary methodological choice.

Importantly, the study does not assume that estimated lead time or sojourn time represents an exact individual-level truth. Instead, these quantities are treated as probabilistic corrections intended to reduce systematic bias at the population level. Within this context, the modeling strategy enhances the validity and interpretability of survival comparisons by addressing a known source of distortion, rather than introducing undue uncertainty. Overall, this integrated approach strengthens confidence in the study's conclusions and supports the use of lead-time–corrected survival analysis in future lung cancer research.

Conclusion

This study presents a comprehensive evaluation of survival in lung cancer by combining clinical characteristics, tumor burden, treatment intent, and screening related timing effects within a lead-time corrected survival framework. Rather than relying solely on observed survival time, the analysis explicitly addressed lead-time bias using tumor growth–based modeling, allowing a more realistic estimation of prognosis. This approach improves the interpretation of survival outcomes and offers valuable insight into the natural history of lung cancer, particularly in populations where the timing of diagnosis varies substantially.

Tumor burden emerged as the most consistent and robust determinant of survival throughout the analyses. Tumor size showed a clear and independent association with mortality in regression models, Kaplan–Meier analyses, and multivariable Cox models, with even small increases in size translating into a measurable increase in the risk of death. After correcting survival time for lead-time bias, indicators of advanced disease such as thrombosis, diffuse cancer, and metastatic involvement remained strongly associated with poor survival, confirming that their prognostic effect reflects true disease severity rather than earlier detection. Treatment intent also played a major role, with the absence of curative treatment consistently associated with substantially higher mortality, highlighting the importance of disease stage at diagnosis and the possibility of curative intervention.

Survival differences according to smoking status were evident in Kaplan–Meier analyses and persisted after lead-time correction, underscoring the long-term negative impact of tobacco exposure on lung cancer outcomes. However, the loss of statistical significance in multivariable models suggests that the effect of smoking on survival is largely mediated through its influence on tumor burden, disease extent, and treatment pathways. Importantly, the comparison of results before and after lead-time correction demonstrated that uncorrected survival analyses can alter both the magnitude and interpretation of prognostic associations. Overall, this study shows that meaningful assessment of lung cancer survival requires simultaneous consideration of biological factors, disease extent, treatment strategy, and diagnostic timing. Incorporating lead-time

corrected survival analysis leads to more reliable identification of prognostic factors and has important implications for clinical decision-making, evaluation of screening programs, and future epidemiological research.

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Appendix

R Software and Packages

All statistical analyses were conducted using R version 4.5.1 , The following R packages were used:

```
library(summarytools) # for descriptive statistics
library(ggplot2)      # for data visualization
library(epitools)     # for epidemiological calculations
library(MASS)         # for statistical modeling
library(survival)     # for survival analysis
library(survminer)    # for enhanced survival plots
```

Appendix Table A1. Multivariable Cox regression using observed survival time

term	HR	CI_lower	CI_upper	p_value
GenderMale	0.964	0.710	1.308	0.812
Age	1.010	0.992	1.028	0.294
AgeGroup> 70 years	0.967	0.677	1.380	0.852
SmokerSmoker related cancer (Groupe A)	1.215	0.882	1.674	0.233
HBVYes	1.224	0.774	1.934	0.387
HIVYes	0.817	0.525	1.272	0.371
TumorSize	1.002	1.001	1.003	0.003
DiabetesYes	0.898	0.714	1.130	0.360
Alcohol_consumptionYes	0.855	0.680	1.074	0.178
ThrombosisYes	2.011	1.577	2.565	0.000
DifuseCancerYes	1.465	1.037	2.071	0.030
MetastaticCancerYes	1.832	1.343	2.499	0.000
CurativeTreatmentTreatment without a curative intent	3.477	2.536	4.768	0.000
Treatment1Yes	1.399	0.866	2.259	0.170
Treatment2Yes	1.470	0.933	2.315	0.096
Treatment3Yes	1.220	0.798	1.865	0.358
Treatment4Yes	1.332	0.887	2.002	0.167
Treatment5Yes	1.192	0.813	1.749	0.368
Treatment6Yes	1.299	0.840	2.008	0.239

Appendix Table A2. Multivariable Cox regression using lead-time–corrected survival

term	HR	CI_lower	CI_upper	p_value
GenderMale	1.229	0.796	1.897	0.352
Age	0.991	0.966	1.018	0.521
AgeGroup> 70 years	1.106	0.667	1.832	0.697
SmokerSmoker related cancer (Groupe A)	0.921	0.604	1.404	0.701
HBVYes	0.670	0.354	1.266	0.217
HIVYes	0.614	0.344	1.098	0.100
TumorSize	1.003	1.001	1.005	0.002
DiabetesYes	0.777	0.564	1.070	0.122
Alcohol_consumptionYes	0.827	0.605	1.131	0.234
ThrombosisYes	1.595	1.098	2.316	0.014
DifuseCancerYes	1.253	0.700	2.242	0.447
MetastaticCancerYes	2.952	1.781	4.892	0.000
CurativeTreatmentTreatment without a curative intent	2.976	2.050	4.320	0.000
Treatment1Yes	1.834	0.950	3.541	0.071
Treatment2Yes	2.239	1.219	4.111	0.009
Treatment3Yes	1.392	0.770	2.517	0.273
Treatment4Yes	1.895	1.106	3.246	0.020
Treatment5Yes	1.504	0.891	2.540	0.127
Treatment6Yes	1.672	0.914	3.061	0.095